

Telomerase

Assistant Lecturer

Hazem Abbas Al-Bustany

M.Sc. in Molecular Biology

hazem.albustany@hmu.edu.krd



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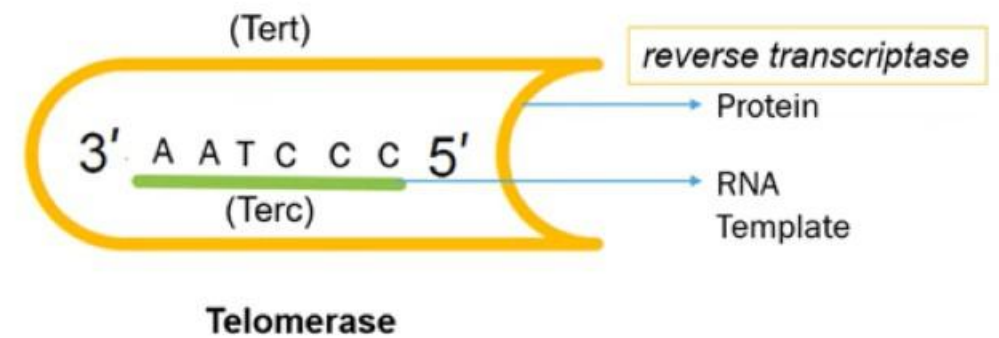
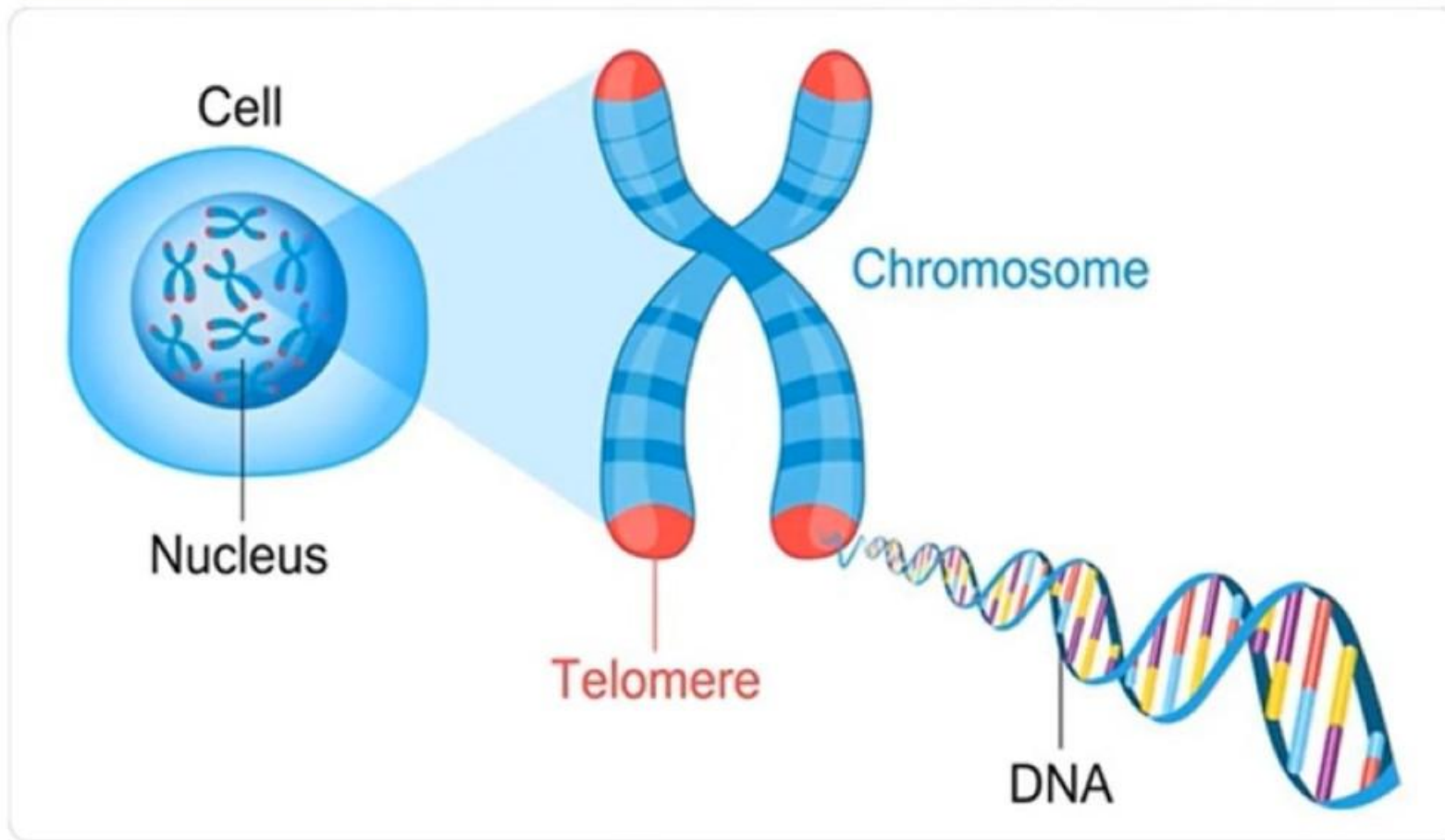
TELOMERASE

Telomerase is a specialized ribonucleoprotein enzyme responsible for maintaining the length and integrity of telomeres, which are repetitive nucleotide sequences located at the ends of linear chromosomes. Telomeres play a crucial role in preserving genomic stability by protecting chromosome ends from degradation, end-to-end fusion, and inappropriate DNA repair.



TELOMERASE

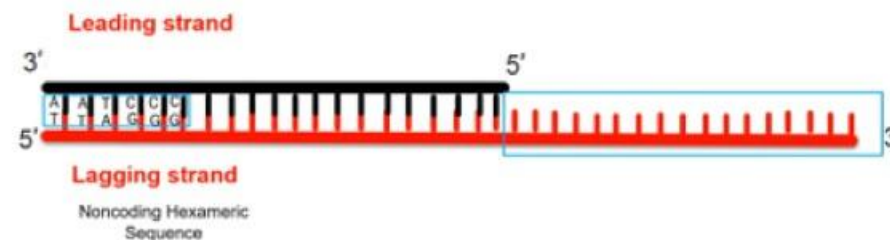
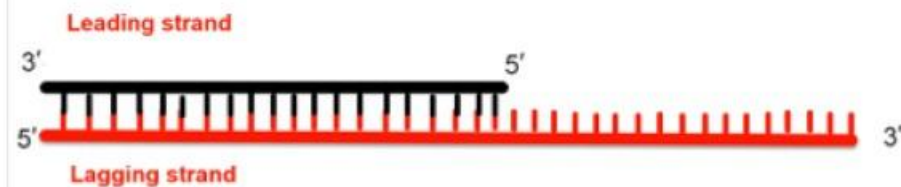
Progressive telomere shortening occurs with each round of cell division due to the end-replication problem, ultimately leading to cellular senescence or apoptosis. Telomerase counteracts this process in specific cell types, thereby influencing cellular lifespan, aging, and disease.



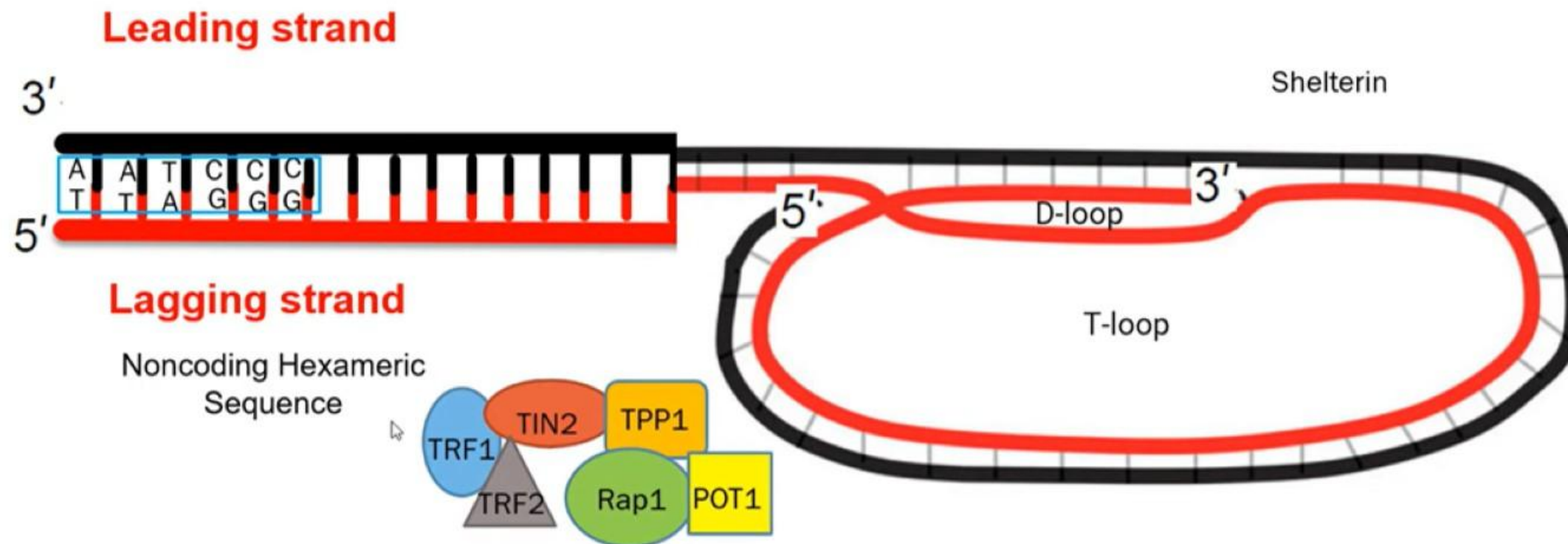
TELOMERE STRUCTURE AND THE END-REPLICATION PROBLEM

Human telomeres consist of **tandem repeats** of the hexanucleotide sequence **TTAGGG**, associated with a multiprotein complex known as the **shelterin complex** (TRF1, TRF2, POT1, TIN2, TPP1, and RAP1). This complex stabilizes telomeres and prevents them from being recognized as DNA double-strand breaks.

During DNA replication, DNA polymerase is unable to fully replicate the **3' end of the lagging strand**, resulting in the gradual shortening of telomeres with each cell division. This phenomenon, termed the **end-replication problem**, limits the proliferative capacity of most somatic cells and underlies the concept of replicative aging.



It prevents the **telomere** ends from being recognized as break points by the DNA repair system.





TRF1 & TRF2 (Telomere Repeat Binding Factors 1 & 2): Directly bind to the double-stranded telomeric DNA (**TTAGGG** repeats) and anchor the shelterin complex to the chromosome ends.

POT1 (Protection Of Telomeres 1): Binds to the single-stranded overhang at the telomere, recruiting TPP1.

TIN2 (TRF1-Interacting Nuclear Protein 2): Acts as a central scaffold, linking TRF1 and TRF2 to the TPP1-POT1 module.

TPP1 (TIN2-Interacting Protein 1): Partners with POT1, forming a module that interacts with TIN2, bringing POT1 to the telomere.

RAP1 (Repressor/Activator Protein 1): Associates with TRF2, helping to regulate telomere length and function.

Telomerase Structure and Components

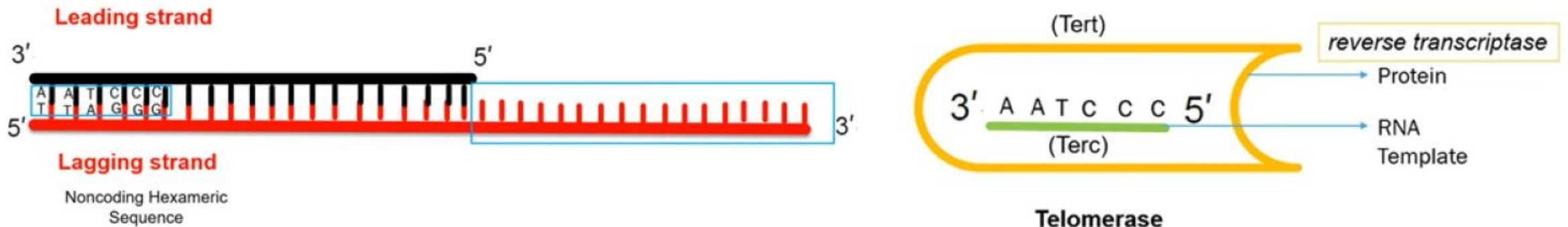
Telomerase is composed of two core components:

- **Telomerase reverse transcriptase (TERT):** A catalytic protein subunit with reverse transcriptase activity.
- **Telomerase RNA component (TERC):** An RNA molecule that serves as a template for the addition of telomeric DNA repeats.

Together, these components enable telomerase to synthesize telomeric DNA de novo, using the RNA template to extend the 3' end of chromosomes. Additional accessory proteins contribute to telomerase stability, localization, and regulation.

Mechanism of Action

Telomerase binds to the single-stranded 3' overhang of telomeric DNA. Using its RNA template, **Telomerase reverse transcriptase (TERT)** catalyzes the addition of **TTAGGG repeats** to the chromosome end. Following nucleotide addition, the enzyme translocates and repeats the process, thereby elongating telomeres. This extension allows conventional DNA polymerases to complete replication of the complementary strand.



Regulation and Physiological Expression

Telomerase activity is tightly regulated and varies among cell types:

- **High activity:** Germ cells, embryonic stem cells
- **Moderate activity:** Adult stem cells
- **Absent or minimal activity:** Most somatic cells

This regulated expression ensures tissue renewal while preventing uncontrolled cell proliferation. Transcriptional regulation of the **TERT gene** is a key determinant of telomerase activity.

Telomerase and Cellular Aging

In somatic cells lacking telomerase activity, progressive telomere shortening eventually triggers DNA damage responses, leading to **replicative senescence** or apoptosis. This phenomenon contributes to organismal aging and limits the regenerative capacity of tissues. The maximum number of divisions a cell can undergo, known as the **Hayflick limit**, is largely determined by telomere length.

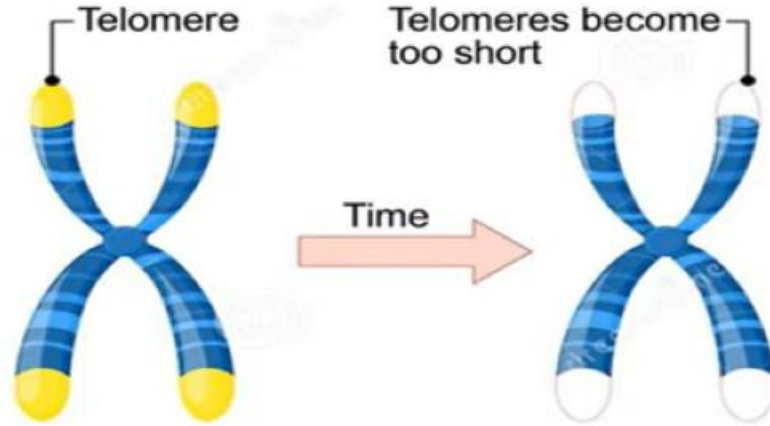


Telomerase in Cancer

Reactivation of telomerase is a hallmark of cancer, observed in approximately **85–90%** of human malignancies. Cancer cells bypass senescence by upregulating telomerase, enabling unlimited proliferation and cellular immortalization. Mutations in the **TERT promoter** are commonly identified in several cancers and lead to increased telomerase expression. Due to its selective activation in tumors, telomerase is a critical target for cancer diagnostics and therapy.

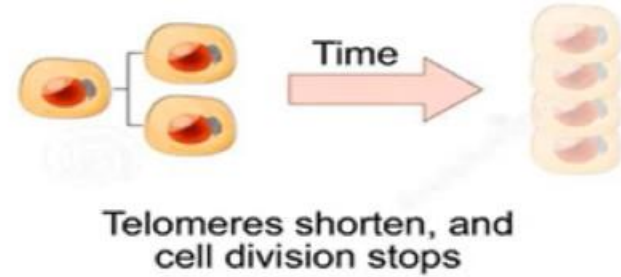
TELOMERE and CANCER

NORMAL CELL

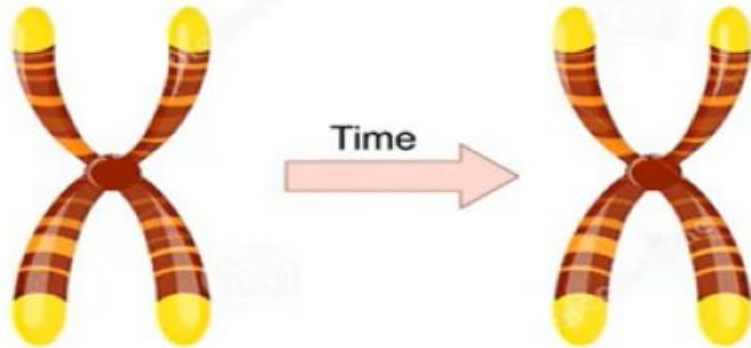


Healthy cell

Cell death

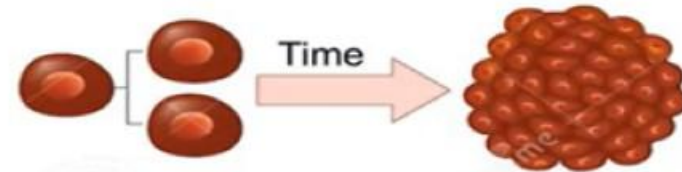


CANCER CELL



Cell with
genetic changes

Malignant
tumour



Telomerase-Related Diseases

Inherited defects in telomerase components result in disorders characterized by premature telomere shortening, including:

- **Dyskeratosis congenita:** A rare inherited disorder caused by mutations in telomere maintenance genes, leading to short telomeres (chromosome caps).
- **Idiopathic pulmonary fibrosis:** A chronic, progressive lung disease where lung tissue becomes thick, stiff, and scarred for unknown (idiopathic) reasons, affecting breathing.
- **Bone marrow failure syndromes:** A group of disorders where the bone marrow can't produce enough healthy blood cells (red, white, platelets).

These conditions highlight the essential role of telomerase in tissue maintenance and hematopoiesis.

Clinical Applications and Therapeutic Targeting

Telomerase is exploited clinically in several ways:

- **Diagnostic marker:** Detection of telomerase activity using the **TRAP assay**
- **Cancer therapy:** Telomerase inhibitors (e.g., imetelstat)
- **Immunotherapy:** Telomerase-based cancer vaccines
- **Regenerative medicine:** Experimental approaches to enhance tissue repair

However, therapeutic inhibition of telomerase may affect normal stem cells, raising concerns regarding toxicity.



The **TRAP** (Telomeric Repeat Amplification Protocol) assay is a sensitive, PCR-based molecular biology technique used to detect and measure the activity of the enzyme telomerase in cell or tissue samples, crucial for understanding cell immortalization, cancer development, and aging. It works by allowing telomerase to add telomeric repeats (**TTAGGG**) to a synthetic oligonucleotide substrate, which is then amplified by PCR and analyzed, typically on a gel, to visualize the characteristic ladder pattern indicating enzyme activity, acting as a biomarker for malignancy.



Conclusion

Telomerase is a fundamental enzyme at the intersection of molecular biology, aging, and cancer. Its role in maintaining chromosomal stability is essential for normal development, while its dysregulation contributes to malignancy and degenerative diseases. Understanding telomerase biology is therefore critical for medical students, clinicians, and researchers, offering insights into both disease mechanisms and potential therapeutic strategies.



Which telomeric DNA sequence is repeated in human chromosomes?

- A. AATTCC
- B. TTAGGG**
- C. CCGGAA
- D. ATCGAT
- E. GGGCCC

The TRAP assay is used to:

- A. Measure telomere length
- B. Measure DNA replication rate
- C. Identify TERT mutations
- D. Detect telomerase activity**
- E. Detect chromosomal translocations

Thank You